SCHOOL OF ENGINEERING SCIENCE AND TECHNOLOGY

DEPARTMENT OF ICT & ELECTRONICS

Fourth Year Project Documentation Guide 2021

Prepared by F.J Kiwa

cut4thproject2023@gmail.com

Note: Avoid contractions (isn't, it's, etc)

Use past tense in your descriptions of all the write ups

Project Documentation Layout and Contents

Title/Front Page

Include the University name, title, author, dissertation supervisor, place, and date

Acknowledgments

Dedication

Release Form

Approval From

Antiplagiarism Report

See CUT policy

Abstract

Give an abstract of the project, which is a summary of the proposal.

Briefly state the (1) research problem, (2) methodology, (3) key results, and (4) conclusion.

Generally, abstracts are half page long (150-200 words) and should be single line spaced.

Example of Abstract Template

Due to **<motivation and background>**, this research investigates **<problem statement>** since **<literature review result>**. In attaining **<aim and objectives>**, this paper utilises **<methodology>** to realise **<results and significance>**. The results enjoys the **<benefit>** but suffers **<limitations>** which is outweighed by <main contribution>, which brings <wider significance>.

Table of Contents

List the key subject headings and subheadings of your thesis with their page numbers. Number the front-matter section in lowercase roman numerals. Be sure to list acknowledgments, appendixes, and bibliography

List of Figures

Include the figure numbers, figure titles, and page numbers.

List of Tables

Include the table numbers, table titles, and page numbers.

CHAPTER ONE- INTRODUCTION

Chapter One constitutes of all sections written in past tense on what you have accomplished. This chapter excludes Literature Review and Methodology chapters. It has the following Sections

1.0 Introduction

1.1 Background of the Study

1.2 Statement of the Problem/ Problem Statement

1.3 Project Aims and Objectives

1.4 Significance of the Study

1.5 Limitations and Delimitations of the Study

1.6 Definition of Terms List unfamiliar terms, acronyms and their meanings

1.7 Conclusion

CHAPTER TWO- LITERATURE REVIEW/ RELATED WORK

2.0 What is a literature review?

A literature review is a survey of scholarly sources that provides an overview of a particular topic. Literature reviews are a collection of the most relevant and significant publications regarding that topic in order to provide a comprehensive look at what has been said on the topic and by whom. The basic components of a literature review include:

- a description of the publication;
- a summary of the publication's main points;
- a discussion of gaps in research;
- An evaluation of the publication's contribution to the topic.

2.1 What is the purpose of a literature review?

The purpose of a literature review is to provide a review of writings on the given topic in order to establish the reviewer's own position in the existing field of scholarship on that topic. A literature review provides a reader with a comprehensive look at previous discussions prior to the one the reviewer will be making in his/her own research paper, thesis, or dissertation.

A literature review can be just a simple summary of the sources, but it usually has an organizational pattern and combines both summary and synthesis. A summary is a recap of the important information of the source, but a synthesis is a re-organization, or a reshuffling, of that information. It might give a new interpretation of old material or combine new with old interpretations.

Depending on the situation, the literature review may evaluate the sources and advise the reader on the most pertinent or relevant. In short, a literature review shows readers where the reviewer is entering the academic conversation on a particular topic in the context of existing scholarship.

2.2 How do I create a literature review?

The length and depth of your literature review depends on the length of your project. If you are writing a 10-page argument paper, you may have room to include 5-6 sources to review, because you will also be establishing your argument as well, but there's no hard equation for how many or how much. Use your judgment and look for other literature reviews in your area

of interest or in the discipline and read, them to get a sense of the types of themes you might want to look for in your own research or ways to organize your final review. Here is a systematic approach to drafting your literature review:

2.2.1 Define Your Goal

If you are writing an argument paper, create a thesis statement with a clear position. If you are evaluating scientific theories, develop a hypothesis to examine. If you are providing a self-contained review of writings on a topic, state your project's purpose. At the beginning of any paper, define your paper's purpose so that the literature review is anchored to a specific point of view.

2.2.2 Do Your Research

Review a number of texts that most closely pertain to your topic and position, and are written by relevant scholars. Understand who the top voices are in your topic's academic field, and be sure to include the most pertinent publications by those scholars.

2.2.3 Ground Summary in Relevance

As you summarize each publication, provide the context for that publication's importance by tying its main points to your thesis, hypothesis, or project statement. How does it relate? Establish its relevance to the discussion.

2.2.4 Develop Review Logically

Think of your literature review as a development of an argument—what were the earliest ideas on the topic and how did they grow and evolve in the academic conversation of these publications? First things first.

2.2.5 Include References/Works Cited List

As you are writing the literature review you will mention the author names and the publication years in your text, but you will still need to compile comprehensive citations for each entry at the end of your review. Follow the APA referencing stttyle as your course requires.

2.3 Output

2.3.1 The Gap

The **gap,** also considered the missing piece or pieces in the research literature, is the area that has not yet been explored or is under-explored. A **gap** is something that remains to be done or learned in an area of research; it's a gap in the knowledge of the scientists in the field of research of your study.

Every research project must, in some way, address a gap–that is, attempt to fill in some piece of information missing in the scientific literature. Otherwise, it is not novel research and is therefore not contributing to the overall goals of science.

2.3.2 Related methodology

It could be introducing a new method that needs to be tested.

2.3.3 Description of a conceptual/proposed solution

Once you identify the gap in the literature, you must tell your audience *how* you attempt to at least somewhat address in your project this lack of knowledge or understanding .

2.4 Sources:

- Use APA Referencing Style
- Literature review must have a minimum of 5 but not more than 10 pages

https://writingcenter.unc.edu/tips-and-tools/literature-reviews/
https://writingcenter.ashford.edu/writing-literature-review

CHAPTER THREE- METHODOLOGY

The methodology is the general strategy that outlines the way in which the project is to be undertaken and, among other things, identifies the methods to be used in it.

3.0 What is the difference between research method and research methodology?

3.0.1 What is a research methodology?

A **methodology** is a system of methods used in a particular area of study or activity. Or the principles and procedures of inquiry in a particular discipline

A **research methodology** is thus the specific procedures or techniques used to identify, select, process, and analyse information about a topic. In a **research** paper, the **methodology** section allows the reader to critically evaluate a **study's** overall validity and reliability

- What kind of methodology have you used and why?
- How have you actually done the project?
- Which data collection methods/sources or tools have you utilized?

"**Research methodology** is a systematic way to solve a problem. It is a science of studying how research is to be carried out. Essentially, the procedures by which researchers go about their work of describing, explaining and predicting phenomena are called research methodology. It is also defined as the study of methods by which knowledge is gained. Its aim is to give the work plan of research."

3.0.2 What is a research method?

Research methods are specific procedures for collecting and analysing data. Developing your research methods is an integral part of your research design. Research method is simply a research tool or a component of research such as interviews. Methodology is the justification for using a particular research method.

The research methods available to an IT researcher fall into two broad categories being quantitative or qualitative.

"**Research methods** are the various procedures, schemes, steps and algorithms used in research. All the methods used by a researcher during a research study are termed as research methods. They are essentially planned, scientific and value-neutral. They include observations, theoretical procedures, experimental studies, numerical schemes, statistical approaches, etc. Research methods help us collect samples, data and find a solution to a problem. Particularly, business and scientific research methods call for explanations based on collected facts, measurements and observations and not on reasoning alone. They accept only those explanations which can be verified by experiments"

3.0.3 Qualitative methods:

- These involve designs, techniques and measurements that do not produce discrete type of data.
- The most commonly utilized methods in this category are
  - ❑ Observations,
  - ❑ Interviews
  - ❑ Participation.
  - ❑ Ethnography,
  - ❑ Case study,
  - ❑ Action research
- These methods permit the research to go beyond statistical results.
- Some of the methods rely on fieldwork.

3.0.4 Quantitative methods

- Involve designs, techniques and measurements that do produce discrete values or quantifiable data.
- The methods in this category require a good amount of time in planning.
- Methods that fall in this category are inclusive of
  - ❑ Experiments
  - ❑ Surveys
- While qualitative methods are based on the naturalistic / interpretivism paradigm, the quantitative methods are based on the positivist paradigm .

3.1 Research Methods

3.1.1 Empirical/Experimental

- The empirical method is a general term for any research method that **draws conclusions from observable evidence.**
- Conduct experiments to test theories. Used in the fields of

    a. artificial neural networks, - implementing new social network using concepts from existing social networks
    b. automating theorem proving,
    c. natural languages,
    d. analysing performances and behaviours
    e. Human Computer Interface
- Experimental evaluation is often divided into two phases.

In an **exploratory phase,** the researcher is taking measurements that will help identify what are the questions that should be asked about the system under evaluation.
Then an **evaluation phase** will attempt to answer these questions.
A well-designed experiment will start with a list of the questions that the experiment is expected to answer Experiments can be performed to study the behaviour of natural systems (e.g.,physics), humans and organizations (e.g., experimental psychology), or artefacts (e.g., performance evaluation of different algorithms).

3.1.2. Computer Simulation Method

This method is normally used for complex phenomena that cannot be implemented in laboratories or maybe too expensive (resource wise). Uses computer simulations to address question difficult to answer in the real application. To investigate systems that are outside of the experimental domain fields:

a) Computer networks e.g. Developing a new protocol.
b) Computer hardware
c) virtual reality

3.1.3 Theoretical Method/ Formal methods and mathematical proof:

- Based on logic and mathematics.
- Output is data models (conceptual model) and Algorithms.
- Uses formal mathematical methods to prove a system has given properties, or to design a system which has those properties main techniques when dealing with problems are *iteration, recursion and induction. For example* to develop a new classifier in AI by using the mathematical representation and theoretical method, you need to prove the efficiency of this model.
- Mathematical models are often used to describe the behaviour of systems.
- This method is widely used in theoretical computer science research such as **developing new logic formalism** and **proving properties of computational complexity**

3.1.4 Design Science

- Methods suitable for software engineering (designing software). Goal is to produce some artefacts. A software system, a data set etc

- Constructive Research – research based upon theories, to test theories.
- A "build" research methodology consists of building an artifact — either a **physical** artifact or a **software** system — to demonstrate that it is possible.
- To be considered research, the construction of the artifact must be **new** or it must include **new features** that have **not been demonstrated before** in other artifacts.
- There is an increasing need for better design of information systems as many organizations have experienced failed IT projects and the adverse effects of bad data.
- A systematic study of design science has been called for in the information systems community.
- With an artefact-centric view of design science - as more artefacts are created, it is important that they are evaluated using appropriate frameworks.
- Design science is an outcome based information technology research methodology, which offers specific guidelines for evaluation and iteration within research projects.

3.1.5 Artificial Intelligence

- The field of artificial intelligence was established more than fifty years ago and has developed a set of methods that are useful for data quality research.
- Agent technologies can be used to automate many tasks such as source selection, data conversion, predictive searches, and inputs that enhance system performance and user experience.

3.1.6 Data Mining

- Evolving out of machine learning or artificial intelligence and statistical learning of statistics, data mining is the science of extracting implicit, previously unknown, and potentially useful information from large datasets.
- The data mining approach can be used to address several data quality issues. For example, data anomaly (e.g., outlier) detection algorithms can be used for data quality monitoring, data cleansing, and intrusion detection
- Data mining has also been used in schema matching to find 1-to-1 matches as well as complex matching relationships.
- While many data mining algorithms are robust, special treatment is sometimes necessary when mining data with certain known data quality issues.

3.1.7 System Design, Implementation

- This research method draws upon design methodology in software engineering, database design and data modelling, and system architecture.
- Researchers often use this method to design and implement proof-of-concept systems.

3.1.8 Action Research

- It is conducted with the primary intention of solving a specific, immediate and concrete problem in the local setting.
- It is hardly concerned with whether the results are generalized to any other setting since its major goal is to seek a solution to a given problem.
- This research approach aims to intervene in the studied circumstances by improving the situation.
- The interest here is in developing an innovation to solve a problem in society with little interest in generalization.
- Action research does develop solutions that are of practical value to the people or organizations with whom the researcher is working with as it is an interactive process while at the same time having minimal contribution to knowledge.

- Though it has limited contribution to theory, it is useful because of its ability to provide answers to problems that cannot wait for theoretical solutions.
- This research approach is also distinguished by there being no control group or variables.
- Its main weakness lies in its potential lack of objectivity on the researcher. The lack of an agreed criterion for evaluating action research is also considered as a weakness.
- Most software application related ideas are developed by collaborative work between the researchers and the beneficiaries

3.1.9 Ethnography

- Its goal is to study a community of people to understand how the members make sense of the social interactions.
- Within Information Technology, its used to understand how technical communities build culture of practices and communication strategies that enable them to collaboratively use technologies innovations.
- It allows Information Technology researchers to have insight into human, social and organizational aspects of Information Technology innovations and applications.
- It is a powerful assessment of technology user's needs and allows for perceiving Information Technology innovations and applications in the eyes of the users.
- It allows the IT researcher to improve the understanding of human thought and action through interpretation of human actions in context .
- The merit of this approach to Information Technologists is its ability to avoid imposing any pre-existing theories as it focuses on how members of a community themselves make sense of their social and cultural settings within the technology environment.
- It involves participation, talking and doing things with participants in their real life situation for a long period of time.
- The main tool of the approach is observation and this is done over a period of time giving it a longitudinal perspective. This approach gives an in-depth knowledge of only a particular context and situation thus making it difficult to generalize.
- It also produces highly qualitative data, which is fairly difficult to present in a manner that is usable to designers and can be too costly if the number of participants were to be increased.

3.1.10 Process

- A process methodology is used to understand the processes used to accomplish tasks in Computing Science.
- This methodology is mostly used in the areas of Software Engineering and Man-Machine Interface which deal with the way humans build and use computer systems.
- The study of processes may also be used to understand cognition in the field of Artificial Intelligence.

3.1.11 Case Study

- The case study is an empirical method that uses a mix of quantitative and qualitative evidence to examine a phenomenon in its real-life context.
- The in-depth inquiry of a single instance or event can lead to a deeper understanding of why and how that event happened.
- Useful hypotheses can be generated and tested using case studies [Flyvbjerg 2006]. This method is widely used in data quality research.

3.1.12 Econometrics

A field in economics, econometrics develops and uses statistical methods to study and elucidate economic principles.

CHAPTER 4- SYSTEM DESIGN AND IMPLEMENTATION

The chapter should describe the design of output, input, database/file and security control specifications. Justify the programming language used. Outline the functional and system requirements. You should also include the system architecture; you can also include the DFDs, flow charts, use case and sequence diagrams depending on what your project is all about.

4.1 Implementation Plan

This is a description of the plan for implementing the project. Here you commit yourself to a course of action and specify the criteria by which your performance is to be judged. Your final grade will depend, in large measure, upon your success in achieving the goals agreed upon between you and your project supervisor.

4.1.1 Components

Deliverable Items

List and describe each of the items you will submit in fulfilment of the project requirements. Deliverable items include, but are not limited to, program executable file(s), program data file(s), program listings, program documentation, user manual and sample program runs.

Milestone Identification

Identify each of the milestones or check points that mark the *completion of some phase* of project implementation. Milestones include, but are not limited to, detailed system analysis, system design, file design, module design, system test design, module coding, system testing and documentation.

Milestone Completion Criteria

List the criteria by which the completion of each milestone is to be judged. If an objective measure is available then it should be specified. If a personal judgment is required then indicate who will make the determination. This information may be given in tabular form if desired.

Schedule of Milestone Completion

Prepare a diagram or table giving the proposed completion date for each of the milestones listed in the previous two sections.

CHAPTER 5- PRESENTATION OF RESULTS

- Reporting the findings without bias
- The results must prove that the problem has been solved. The problem must be tied to the results
- Examples:
- A graph that shows the performance of a coronavirus detecting devices/application over a number of suspected patients.
- A graphical view of how the file storage application is returning search results.
- A table or graph that shows how accurate is the recommendation algorithm in terms of predicting a given sample of people's preferences

CHAPTER 6- SUMMARY, CONCLUSION AND RECOMMENDATION

- Summary
- System maintenance
- Future work

REFERENCE

References at least 10 published articles using Harvard referencing style

USER'S MANUAL

APPENDIXES

.

After you have designed your prototype/ artefact, you need to create a promotional advert

Promotional Advert Design

What Is an Advert?

An advert is a visual presentation comprising whatever the contributor wishes to display on the visual board. An adverts main mandate is to call to attention in the course of speaking or writing with visuals.

Promotional advertisements are short-term marketing strategies, businesses use to boost the sales of a certain product or service. You are to advertise your product or services to boost awareness, increase sales or highlight updates of a new or existing product.

Your advert should not be more than 90seconds!!

Designing your Advert

In preparing an advert, simplicity is the key, so there should be a minimum of clutter and amaximum of concise, informative statements and attractive, enlightening graphics.

An advert should tell a story. As always in a scientific presentation, the broad outline includesa statement of the problem, a description of the method of attack, a presentation of results, and then a summary of the work.

An advert should not contain a lot of details—the presenter can always communicate the finepoints to interested participants

Advert should include all multimedia; multimedia is a system of relaying information or entertainment that includes many different forms of communication. Multimedia might include video, audio clips, animations, text, voice and still photographs.

Attributes of a good advert

Make ‘em laugh

If people come away from your ad laughing, that's a positive association with your product/services. A knee-slapping ad is likely to be shared widely, expanding your reach and providing organic exposure. Try sticking to family-friendly humor and situations to ensure your ad appeals to the broadest possible audience. Think of a humorous ad as an opportunity to share your company's personality.

Keep it simple

You only have a few precious seconds to capture the audience; you'll need to efficiently tell your story and make viewers remember your company. Your ad should quickly set the scene and arrive at the main pitch without confusing viewers or overcomplicating your message.

Stay positive

You do not want viewers to come away from your ad thinking, "Wow, that was a bummer." Even if you handle potentially difficult subjects, like insurance, make it positive and uplifting. Use your ad to show how your company can make the viewer's life easier (even in a time of crisis).

Be relatable

Use your ad to depict your product in real-life situations. Through the ad, viewers can see themselves using your product or service either through relatable characters or relatable situations.

Conversational

Instead of lecturing to your audience about the value of your product/services, have characters talking about what need your company is meeting. Use the conversation to portray how you would like to have your company realistically discussed by customers.

Make it memorable

Some of the most memorable ads have a jingle, tagline or mascot that's permeated pop culture. Whether it is catchy or maybe even obnoxious, there's a hook that makes people think about the ad all day. A hilarious punch line or highly quotable dialogue can be just as effective.

Thought-provoking

An ad is an opportunity to challenge viewers' pre-conceived notions about an issue related to your product or service. If your industry is stereotyped in a certain way, or has a reputation you'd like to challenge, use your ad to show how your product/service is different. There is a fine line between controversial and thought provoking, but it could pay off by having viewers think about your ad long after it is ended.

Create an emotional connection

Beyond being relatable, a great ad builds an emotional connection with the viewer. That emotion could be joy, nostalgia, love or trust, but is ultimately positive. Make viewers feel something when they see your ad, and in turn, associate that emotion with your product/service.

Example of an advert: youtube.com/watch?v=G_UdUgO4DR8

Other examples I will share on the Whatsapp platform

Reference List

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